Improving Probabilistic Diffusion Models with Optimal Diagonal Covariance Matching

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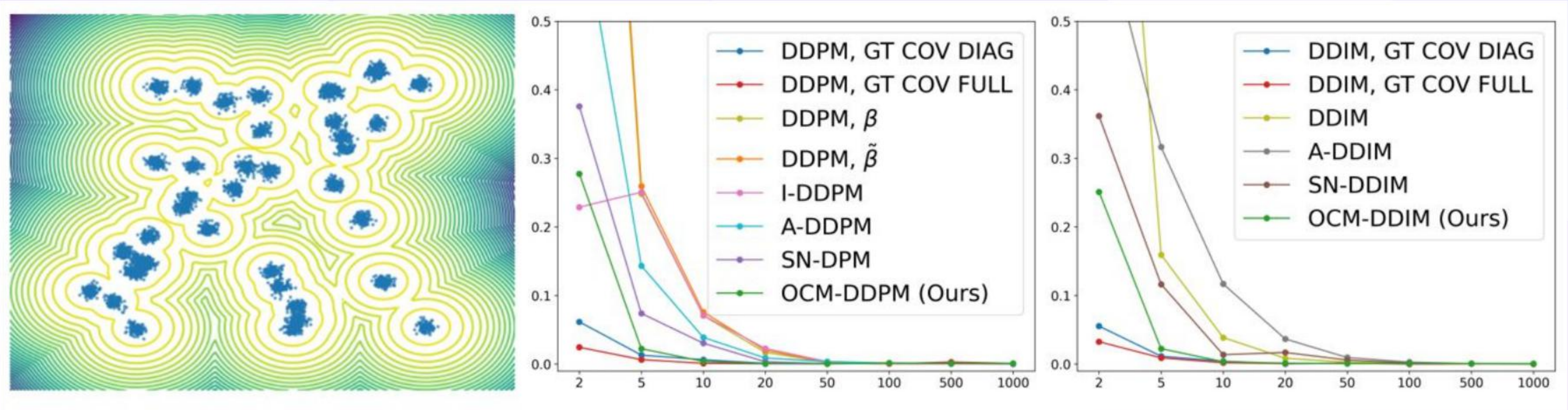
UCL & collaborators · 2025

01 Problem: fixed variance slows diffusion



- Fixed or heuristic variances force very large step counts to reach strong sample quality and likelihood
- Learning state-dependent covariance cuts function evaluations while maintaining fidelity
- Our objective: estimate a better covariance to accelerate sampling without worsening negative log likelihood
- Focus on diagonal covariance for scalability to high-dimensional images

Sohl-Dickstein et al., 2015; Ho et al., 2020; Song & Ermon, 2019; Nichol & Dhariwal, 2021; Bao et al., 2022a; Bao et ...

02 DDPM/DDIM and why covariance matters

Modeling Capability

Covariance Type	+#Passes	Intuition
x_t-independent Isotropic β (Ho et al., 2020)	0	Cov. of $q(x_t\|x_{t-1})$
x_t-independent Isotropic $\tilde{\beta}$ (Ho et al., 2020)	0	Cov. of $q(x_t\|x_{t-1}, x_0)$
x_t-independent Isotropic Estimation (Bao et al., 2022b)	0	Estimate from data
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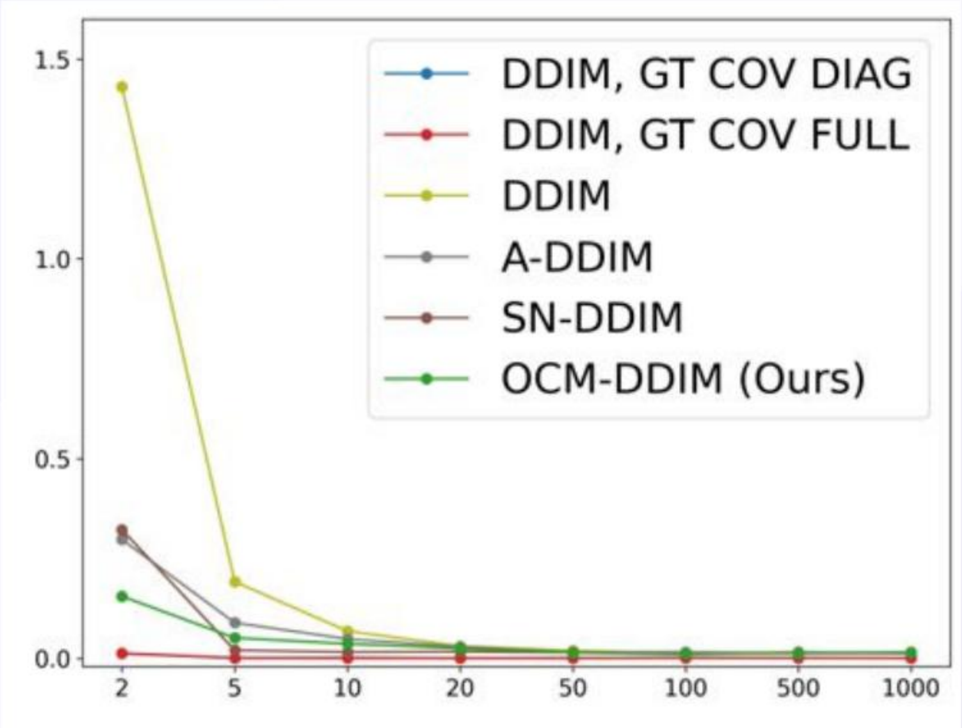
x_t-dependent Diagonal OCM (Ours)	1	Learn from score
x_t-dependent Diagonal Estimation (Zhang et al., 2024b)	$2M$	Estimate from score
x_t-dependent Diagonal Analytic (Zhang et al., 2024b)	D	Calculate from score
x_t-dependent Full Analytic (Zhang et al., 2024b)	D	Calculate from score

- When the step budget is small, the chosen covariance of the denoising distribution strongly impacts outcomes
- Heuristics such as using forward or posterior variances behave similarly only with very large steps
- **Goal**: improve covariance for fewer evaluations in DDPM and DDIM, including the zero-step target in DDIM
- Poor covariance choices degrade likelihood and diversity under aggressive skipping

Nichol & Dhariwal, 2021; Song et al., 2020

03 Optimal covariance from the score

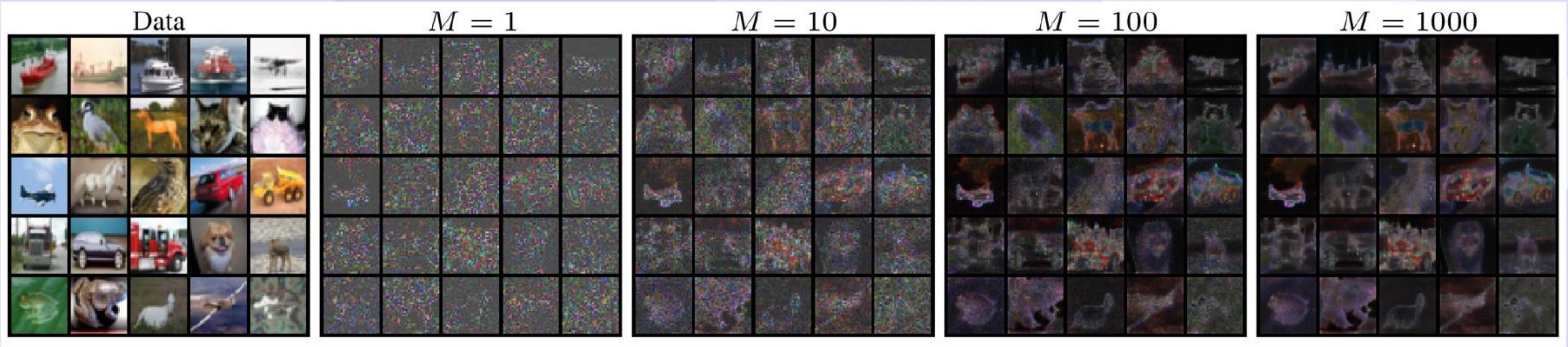
- A generalized covariance identity expresses the optimal posterior covariance as a functional of the score and its Hessian
- With scaled Gaussian convolution, the optimal state-dependent covariance has a closed form from the score
- **Practical choice**: use only the diagonal of the Hessian to gain speedups with strong accuracy
- Links to higher-order Tweedie ideas and optimality under divergence criteria



(d) DDIM MMD v.s. Steps

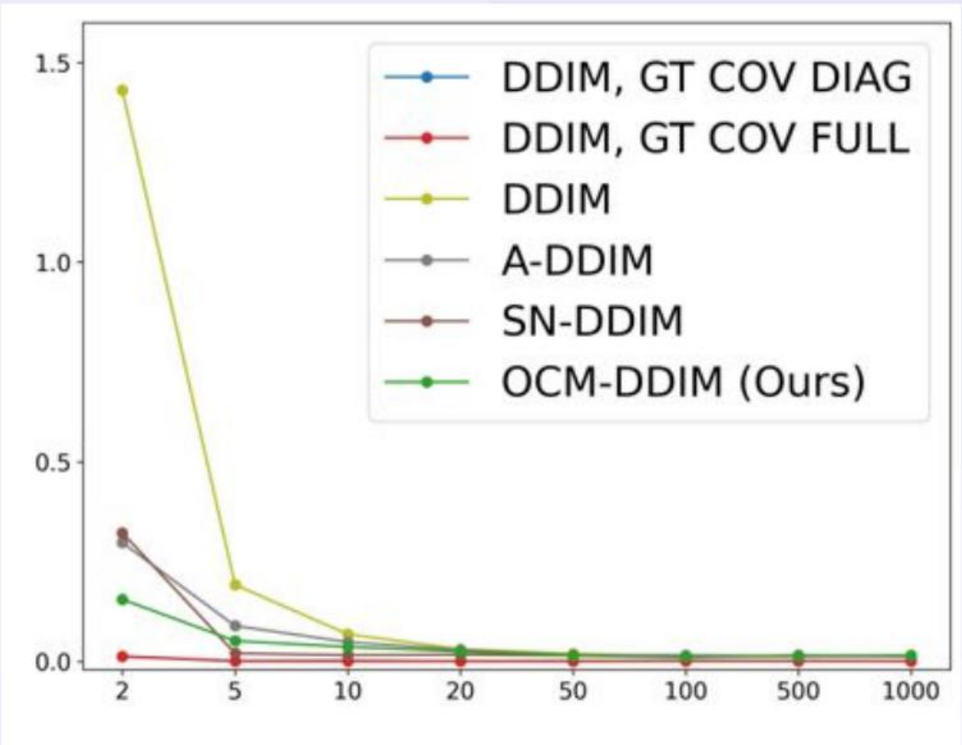
Efron, 2011; Robbins, 1992; Zhang et al., 2024b; Bao et al., 2022b

04 Hessian cost and diagonal estimation



- Exact Hessians demand quadratic memory and many network evaluations per denoising step
- Hutchinson or Rademacher estimators approximate the diagonal but need many probe vectors for low variance
- Large probe counts imply many extra forward and backward passes, negating acceleration
- We need a way to predict the diagonal Hessian with single-pass overhead at inference

05 Unbiased Optimal Covariance Matching (OCM)

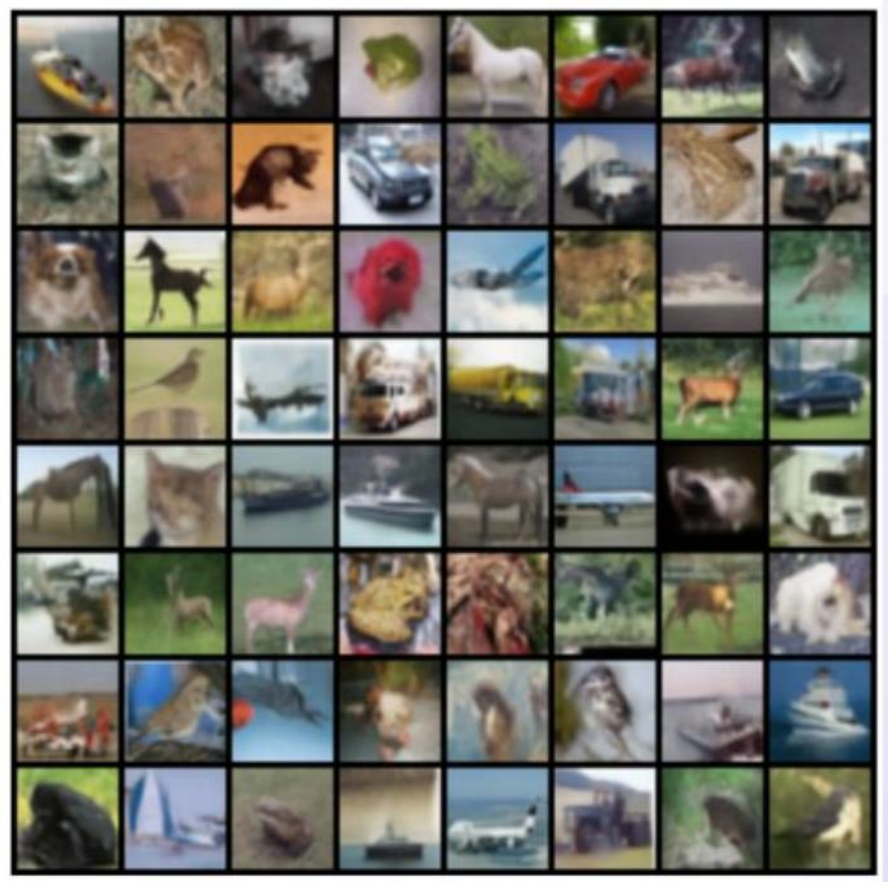


(d) DDIM MMD v.s. Steps

- Train a network to predict the diagonal of the score Hessian using an **unbiased** upper-bounding objective
- Monte Carlo over Rademacher vectors keeps the estimator unbiased and works well even with a single sample
- At inference, form a diagonal covariance from the predicted Hessian diagonal with negligible extra cost
- Reduces covariance error relative to data-only learning objectives

06 Applying OCM to skip-step DDPM/DDIM

- Learn the Hessian-diagonal head using a pretrained score network, then set diagonal covariance at each step
- Skip-step DDPM: parameterize a Gaussian transition with the usual mean and the learned diagonal covariance
- Skip-step DDIM: sample a non-degenerate image estimate with learned covariance, then update the latent state
- Works for both pixel and latent diffusion models

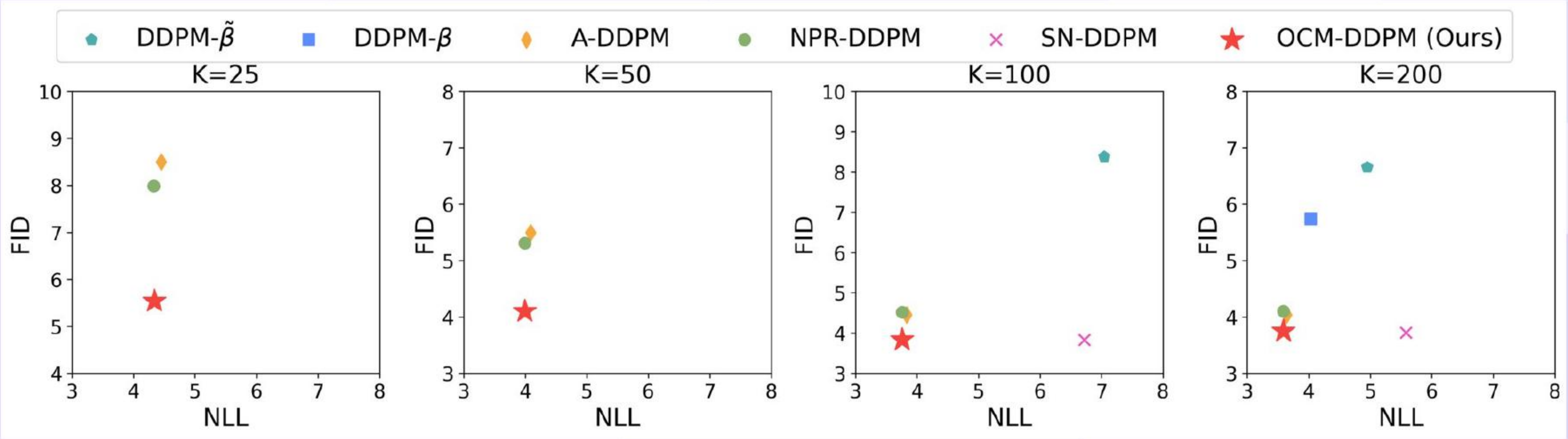


07 Shared-backbone Hessian head

	Score prediction network	Score & SN prediction networks	Score & Diagonal Hessian prediction networks
CIFAR10 (LS)	52.54 M / 44.37 ms	52.55 M / 44.61 ms (+0.5%)	52.55 M / 44.51 ms (+0.3%)
CIFAR10 (CS)	52.54 M / 45.13 ms	52.55 M / 45.24 ms (+0.2%)	52.55 M / 45.23 ms (+0.2%)
CelebA 64x64	78.70 M / 67.88 ms	78.71 M / 69.15 ms (+1.9%)	78.71 M / 68.89 ms (+1.5%)
ImageNet 64x64	121.06 M / 106.58 ms	121.49 M / 112.53 ms (+5.6%)	121.49 M / 112.23 ms (+5.3%)
LSUN Bedroom	113.67 M / 692.58 ms	114.04 M / 771.55 ms (+11.4%)	114.04 M / 770.73 ms (+11.2%)
ImageNet 256x256	675.13 M / 22.84 ms	Missing model	677.80 M / 23.01 ms (+0.7%)

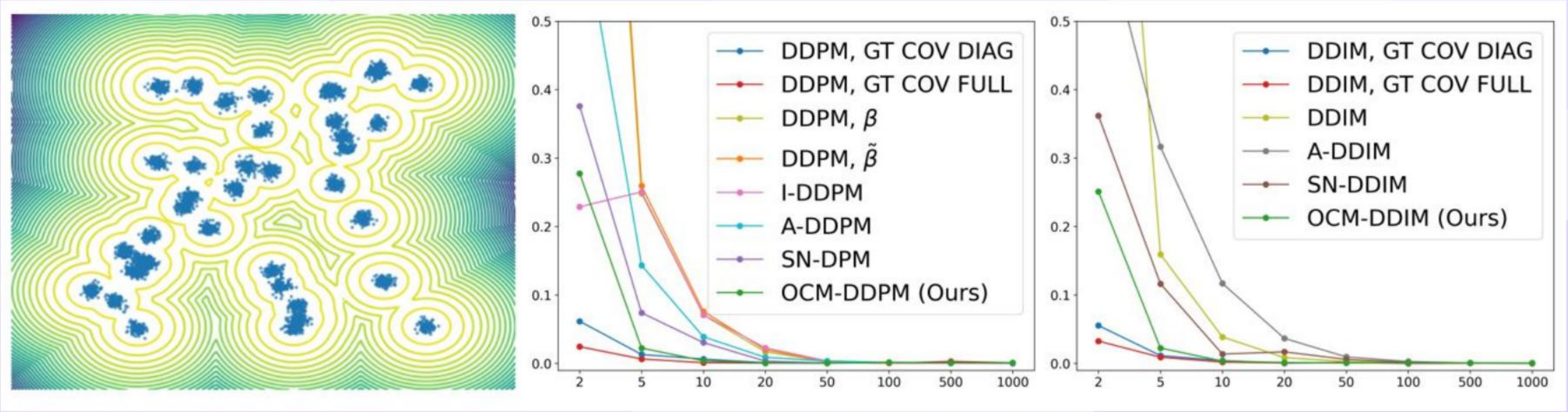
- Attach a lightweight Hessian-diagonal head to shared BaseNet features such as UNet or DiT
- Freeze the pretrained score network and train only the Hessian head with the OCM objective
- Minimal compute and memory overhead; prediction runs in parallel with mean or score inference
- Simple to integrate into existing diffusion training pipelines

Preview: best FID-NLL trade-off



- On CIFAR-10 with the cosine schedule, OCM achieves the best frontier of sample quality versus likelihood
- Better likelihood than heuristic and learned baselines at low evaluation counts
- Consistent gains for both DDPM and DDIM sampling regimes
- Improves quality without sacrificing tractable density estimation

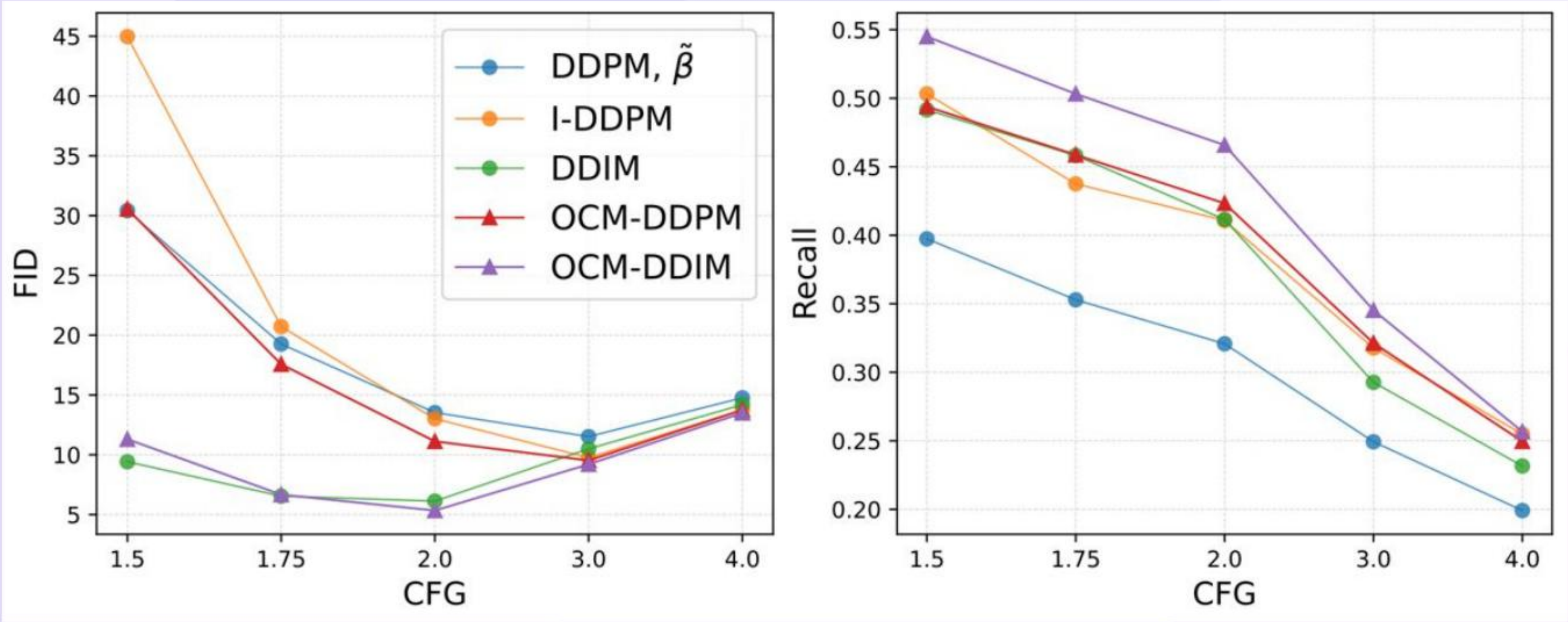
MoG: setup, evaluation, and results



- Two-dimensional mixtures with known scores enable explicit covariance error measurement
- OCM yields lower diagonal-covariance error and lower discrepancy with fewer denoising steps
- Full covariance outperforms diagonal on toys, motivating low-rank or block extensions
- Clear wins over prior diagonal learning approaches across step budgets

Gretton et al., 2012

DiT/CFG on ImageNet 256×256



- With 10 steps and classifier-free guidance near 2.0, OCM attains the best sample quality and higher recall
- Maintains tractable likelihood while improving diversity compared to deterministic-variance samplers
- Mean function from DiT is unchanged; covariance is learned via OCM
- Scales to large images with competitive efficiency

Peebles & Xie, 2023; Sajjadi et al., 2018

FID, steps, diversity

Method	CIFAR10	CelebA 64x64	LSUN Bedroom	ImageNet 256x256
DDPM	90 (6.12)	> 200	130 (6.06)	21 (5.89)
DDIM	30 (5.85)	> 100	Best FID > 6	11 (5.58)
Improved DDPM	45 (5.96)	Missing model	**90** (6.02)	22 (6.08)
Analytic-DPM	25 (5.81)	55 (5.98)	100 (6.05)	Missing model
NPR-DPM	1.002×23 (5.76)	1.013×50 (6.04)	1.021×**90** (6.01)	Missing model
SN-DPM	1.005×17 (5.81)	1.019×22 (5.96)	1.114×92 (6.02)	Missing model
OCM-DPM (ours)	1.003×**16** (5.83)	1.015×**21** (5.94)	1.112×**90** (6.04)	1.007×**10** (5.33)

- Fewest steps to reach an FID near six in most benchmarks after accounting for small head overhead
- Competitive or improved likelihood relative to recent moment-based methods at similar cost
- Better quality–diversity trade-off across CIFAR-10, CelebA, LSUN, and ImageNet
- Only method among strong baselines delivering both likelihood and sample quality consistently

Heusel et al., 2017

12 Covariance choices and applicability

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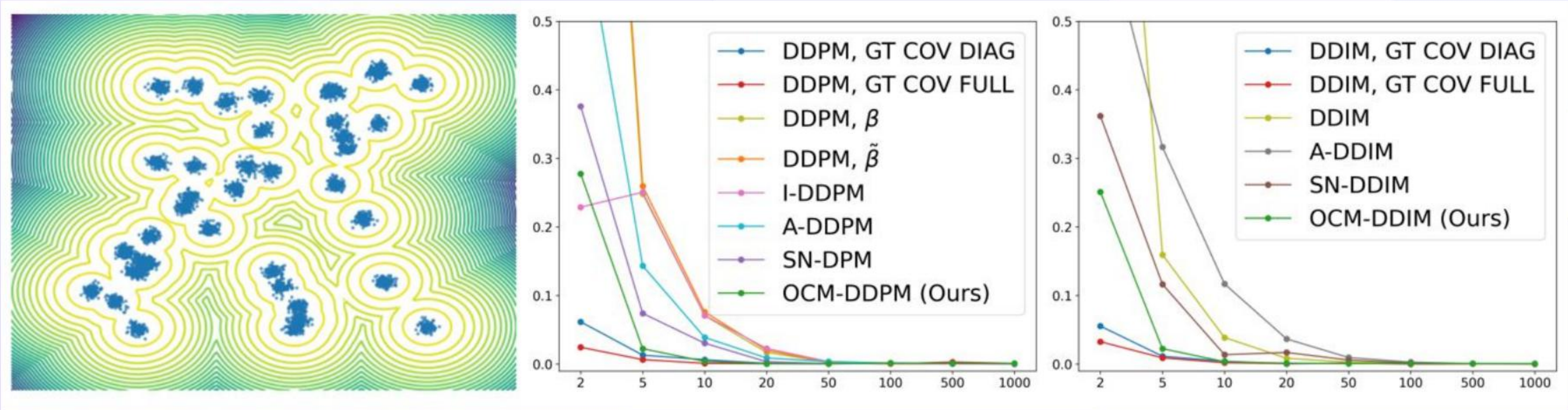
- OCM versus heuristic betas, analytic isotropic, variational lower bound learning, and noise-moment methods
- OCM applies cleanly to DDPM skip steps and to DDIM with zero-step targets, avoiding ill-defined cases
- Uses second-order score information and reduces error amplification near the noise-free limit
- Broadly compatible with pretrained diffusion backbones and schedules

13 Broader acceleration and distillation

	FiD ↓				NLL ↓			
# timesteps	5	10	15	20	5	10	15	20
DDPM, $\tilde{\beta}$	58.28	34.76	24.02	19.00	203.29	74.95	44.94	32.20
DDPM, β	254.07	205.31	149.67	109.81	7.33	6.51	6.06	5.77
OCM-DDPM	**38.88**	**21.60**	**13.35**	**9.75**	**6.82**	**4.98**	**4.62**	**4.43**

- Complementary to fast continuous-time solvers and trainable forward processes such as bridge methods
- Distillation offers few-step generation but often lacks tractable likelihood limiting downstream use
- OCM preserves tractable density, aiding compression and other likelihood-critical tasks
- Can pair with expressive decoders or alternative conditionals for further gains

14 Beyond diagonal; applications



- Diagonal covariance limits fidelity; explore low-rank or block-diagonal structures for better trade-offs
- Applications: scientific modeling, likelihood-based compression, video generation, and inverse problems
- Higher-order moment matching could accelerate distillation and improve stability
- Integrating improved covariance into general stochastic and deterministic solvers is promising

Thank You

Questions & Discussion